

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CHEMISTRY

Paper 3 Free Response

964729/03

21-18 September 20186

2 hours

Formatted: Not Highlight

Additional Materials: Writing Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, ~~highlighters~~, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

Answer any **four** questions.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

[Turn over

This document consists of ~~13~~14 printed pages and ~~10~~ blank page.

Formatted: Left

[Turn over

Section A

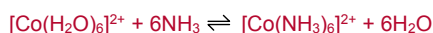
Answer all questions in this section.

Formatted: Font: 11 pt, Complex Script Font: 11 pt

1 Cobalt is a typical transition element which is commonly used as a catalyst and metal for electroplating. Cobalt also forms complex ions with ligands such as H_2O and NH_3 to give various coloured octahedral complexes such as $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Co}(\text{NH}_3)_6]^{2+}$ which are pink and yellow respectively.

(a) The ligand exchange in octahedral complexes is one of the most extensively studied reactions in transition metals.

An example of a ligand exchange reaction involving cobalt(II) ions is:



(i) Explain why cobalt forms coloured complexes. [3]

- The presence of ligands causes the energy level of the five 3d orbitals to be split into two different levels (crystal field splitting).
- The energy difference, ΔE , corresponds to wavelengths in the visible spectrum.
- When light energy is absorbed by the substance, an electron is promoted from a d orbital of lower to one of higher energy (d-d transition)
- Unabsorbed wavelengths are transmitted and the colour of the complex is complementary to the colour absorbed.

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

[3] for 4 marking points

[2] for 3 marking points

[1] for 2 marking points

Formatted: Font: 11 pt, Font color: Red

Formatted: Font: 10.5 pt

(ii) Suggest why $[\text{Co}(\text{NH}_3)_6]^{2+}$ is of a different colour from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$. [1]

- Different ligands give rise to a difference in the splitting of the d-orbitals, cause a different energy gap (ΔE).
- A different wavelengths of visible light is absorbed and thus a different wavelength of light is observed.

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

[1] for 2 marking points

Formatted: Font: 11 pt, Font color: Red

(iii) A student wishes to investigate the kinetics of the ligand exchange reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ to form $[\text{Co}(\text{NH}_3)_6]^{2+}$ by using a spectrometer. This machine measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a few cm^3 of the coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent.

The spectrometer is set to use the wavelength of light that is absorbed most strongly by the complex ion. The amount of light absorbed is expressed as an absorbance value. The more concentrated the solution, the higher the absorbance value. The temperature of the sample in the spectrometer can be thermostatically controlled for reaction rate analysis for which the sample has to be kept at a constant temperature.

Outline the experimental procedure on how the student would accurately determine the initial rate of the ligand exchange reaction at 5°C .

[3]

The details of the use of ~~No details regarding use of specific glassware for measurement are not required.~~ [3]

- ~~Spectrometer is set up to absorb the wavelength of violet (monitoring product) or green (monitoring reactant) and maintain the temperature of the sample at 5 °C.~~
- ~~Separate solutions of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and NH_3 are cooled to 5 °C in an ice-water bath.~~
- ~~Stoichiometric amounts of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and NH_3 (or excess NH_3) are mixed and swirled. A few cm^3 of the coloured solution is quickly placed into the spectrometer.~~
- ~~A stop watch is started.~~
- ~~The concentration of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (or $[\text{Co}(\text{NH}_3)_6]^{2+}$) is then determined by measuring the absorbance of the reaction mixture at time = 0 minute and then at regular time intervals (e.g. every 5 minutes), to obtain at least 5 measurements.~~
- ~~A graph of absorbance value against time is plotted.~~
- ~~The initial rate is found by drawing a tangent at time = 0 minute.~~

Formatted: Font: Bold

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

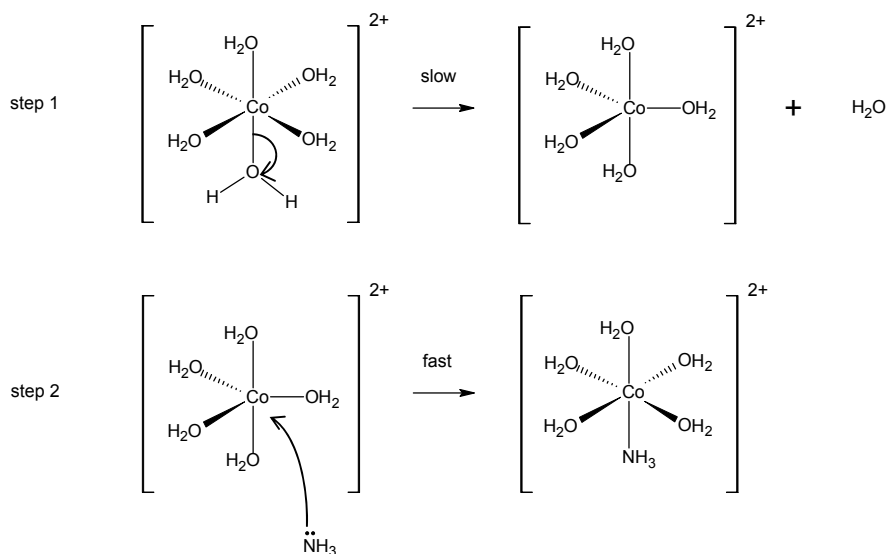
Formatted: Font: 11 pt

7 marking points

[1] for 2 marking points

Formatted: Font: 11 pt, Font color: Red

(iv) When $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is mixed with an excess of $\text{NH}_3(\text{aq})$, each H_2O molecule is replaced by a NH_3 molecule one at a time. Given that the stepwise formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ undergoes a dissociative mechanism which resembles a $\text{S}_{\text{N}}1$ mechanism in organic chemistry. Suggest a possible mechanism for the formation of $[\text{Co}(\text{H}_2\text{O})_5\text{NH}_3]^{2+}$ from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and show clearly how the shape of the complex ion changes. In -your mechanism, show appropriate curly arrows, lone pairs and dipoles.
[3]



Square pyramidal intermediate is accepted as well.

[1] for correct shapes

[1] for arrows and lone pair on NH_3

Formatted: Font color: Red

[1] for intermediate and side product

(v) State the rate equation for the above ligand exchange reaction. [1]



Formatted: Font: 11 pt

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

(vi) Hence, predict and explain the effect on the rate of reaction, if any, when the ammonia ligand is replaced by a fluoride ion.

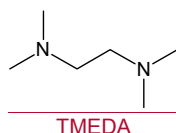
[1]

The rate of reaction remains constant as the rate is independent of the incoming ligand. [1]

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.54 cm

(b) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ can also undergo ligand exchange reactions with TMEDA to form $[\text{Co}(\text{TMEDA})_3]^{2+}$.



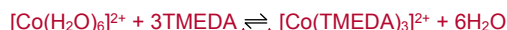
By considering the entropy and enthalpy changes during the formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and that of $[\text{Co}(\text{NH}_3)_6]^{2+}$ from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, suggest how the standard Gibbs free energy change of the two reactions will compare in sign and in magnitude.

Hence, predict which reaction will be the more spontaneous. Explain your reasoning.

[3]



Formatted: Font: 11 pt



Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

ΔH for formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and $[\text{Co}(\text{TMEDA})_3]^{2+}$ is similar in magnitude and sign due to the breaking of 6 similar Co—O bonds and forming of 6 similar Co—N bonds. [1]

ΔS for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ would be more positive than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$ because there is an increase in number of aqueous particles when $[\text{Co}(\text{TMEDA})_3]^{2+}$ is formed, allowing more ways of arranging the particles. [1]

Since ΔH for formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and $[\text{Co}(\text{TMEDA})_3]^{2+}$ are similar in magnitude and sign and ΔS for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ is more positive than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$, ΔG for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ would be more negative than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and hence more spontaneous. [1]

ΔH for formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and $[\text{Co}(\text{TMEDA})_3]^{2+}$ is similar in magnitude due to breaking of similar Co—O bonds and forming of similar Co—N bonds. [1]

Formatted: Font: 11 pt

Formatted: Justified, Indent: Left: 1.27 cm

Formatted: Font: 11 pt

Formatted: Justified

ΔS for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ would be more positive than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$ because there is an increase in number of aqueous particles as $[\text{Co}(\text{TMEDA})_3]^{2+}$ is formed, allowing more ways of arranging the particles. [1]

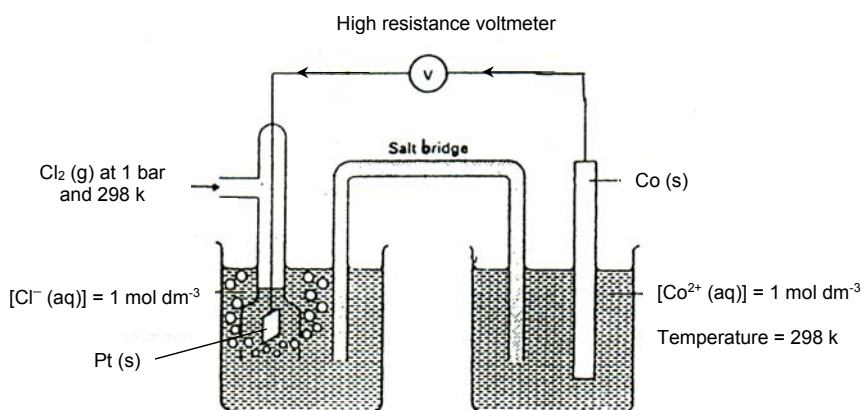
Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Justified, Indent: Left: 1.27 cm

Since ΔH for formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and $[\text{Co}(\text{TMEDA})_3]^{2+}$ are similar in magnitude and ΔS for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ is more positive than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$, ΔG for formation of $[\text{Co}(\text{TMEDA})_3]^{2+}$ would be more negative than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$ and hence more spontaneous. [1]

- (c) (i) Draw a fully labelled diagram of an electrochemical cell composed of a standard Cl_2/Cl^- electrode and a standard Co^{2+}/Co electrode. Indicate the direction of the electron flow. [3]



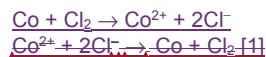
[1] correct set-up

[1] concentration of ions, 298 and 1 bar

[1] electron flow

- (ii) Calculate the $E^\ominus_{\text{cell}}$ of the electrochemical cell and write a balanced equation for the cell reaction. [42]

$$E_{\text{cell}} = +1.36 - (-0.28) = +1.64 \text{ V} [1]$$



- (iii) Using your answer in (ii), calculate ΔG for the cell reaction. [1]

Overall 2 mol of electrons are transferred.

Formatted: Justified
Formatted: Font: 11 pt
Formatted: Justified, Indent: Left: 1.27 cm
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt

Formatted: Font: 11 pt
Formatted: Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm
Formatted: Font: 11 pt, Italic
Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm
Formatted: Font: 10.5 pt
Formatted: Font: 11 pt
Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm
Formatted: Font: 11 pt
Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Font: 11 pt
Formatted: Complex Script Font: 11 pt
Formatted: Font: 11 pt, Complex Script Font: 11 pt
Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

$$\Delta G = -nFE_{\text{cell}} = -2 \times 96500 \times 1.64 = -317\,000 \text{ J mol}^{-1} = -317 \text{ kJ mol}^{-1}$$

- (iv) Use the Data Booklet to suggest the effect on the cell potential of this cell of adding excess aqueous ammonia to the $\text{Co}^{2+}|\text{Co}$ half cell. Explain your answer. [1]



In the presence of ammonia ligands, Co^{2+} undergoes ligand exchange to form a more stable complex ion of $[\text{Co}(\text{NH}_3)_6]^{2+}$. The $E^\ominus(\text{Co}^{2+}|\text{Co})$ becomes more negative and the E_{cell} becomes more positive.

Or

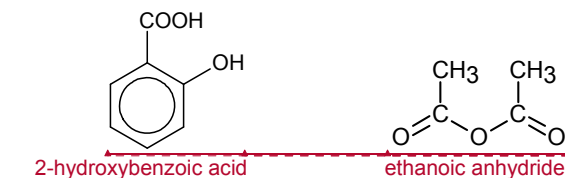
The formation of $[\text{Co}(\text{NH}_3)_6]^{2+}$ decreases the concentration of Co^{2+} . This caused the position of equilibrium for $\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$ to shift left. The $E^\ominus(\text{Co}^{2+}|\text{Co})$ becomes more negative and the E_{cell} becomes more positive.

[Total: 24/2]

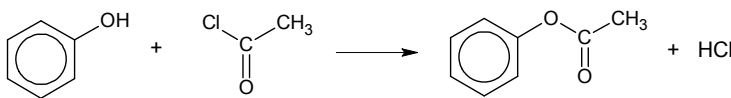
Answer any four questions.

- 12(a) Aspirin is one of the most widely used drug in the world. It is a powerful analgesic (pain-reliever), antipyretic (fever reducer) and anti-inflammatory drug.

It is synthesised using 2-hydroxybenzoic acid and ethanoic anhydride. 8 – 10 drops of 85% phosphoric acid which catalyses the reaction is added. The reaction mixture is then heated under reflux for around fifteen minutes. The other product of this reaction is ethanoic acid.

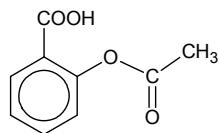


Ethanoyl chloride and phenol can undergo condensation reaction.

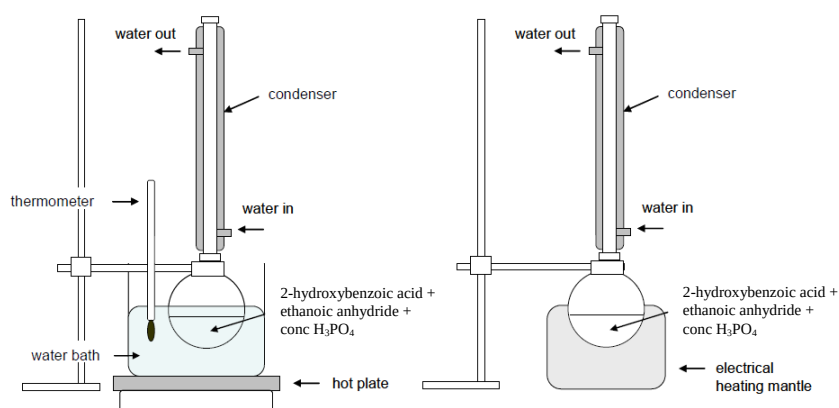


Ethanoic anhydride and 2-hydroxybenzoic acid can undergo a similar reaction to form aspirin.

- (i) Draw the structure of aspirin. [1]



(ii) Draw a labelled diagram of the assembled apparatus for the synthesis of aspirin. [3]



[1] flask fitted with Liebig condenser

[1] clear label of water entering and leaving the condenser in a correct manner

[1] correct heat source (thermostated water bath if thermometer is not drawn)

The mechanism for the reaction between 2-hydroxybenzoic acid and ethanoic anhydride involves four steps. It is proposed as below:

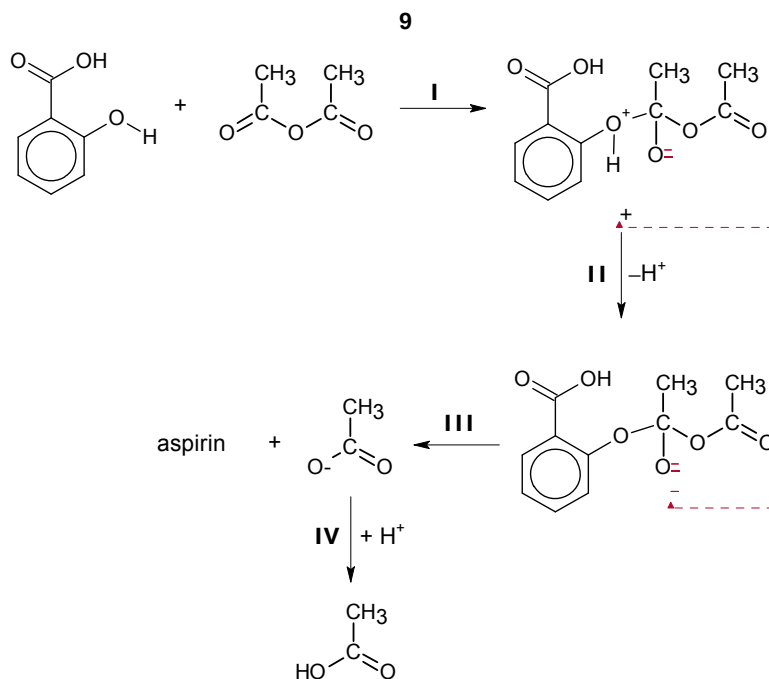
Formatted: Left, Indent: First line: 1.27 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt



Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: Bold

Formatted: Font: (Default) Arial, Bold, Complex Script Font: Arial

Formatted: Font: Bold

(iii) Using the information given above, state the type of mechanism in step I. [1]

Nucleophilic addition [1]

Formatted: Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 3 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Font: 11 pt

(iv) Copy and complete the whole mechanism above by showing any relevant charges, lone pairs of electrons and movement of electrons in your answer. [3]

Formatted: Right: 0 cm, Space After: 10 pt, Don't add space between paragraphs of the same style, Line spacing: Multiple 1.15 li, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 3 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

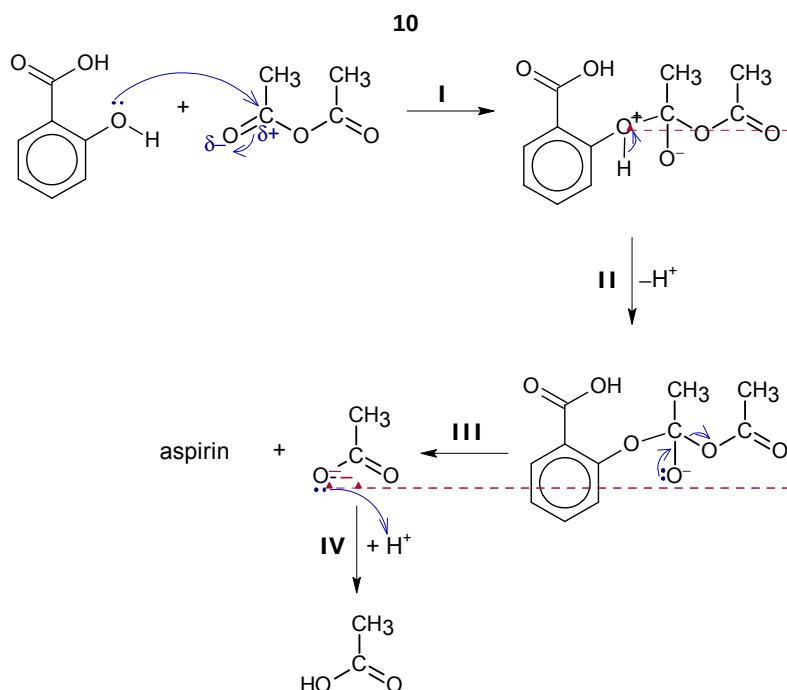
Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.54 cm, No bullets or numbering



Formatted: Font: Bold

Formatted: Font: Bold

Formatted: Font: Bold, Not Superscript/ Subscript

4 steps: 3 marks; 3 steps: 2 mark; 2 steps: 1 mark

(v) State a reason why ethanoic anhydride is used rather than ethanoyl chloride for the synthesis of aspirin. [1]

Safer because ethanoic anhydride is less corrosive as it does not produce corrosive and poisoning / toxic fumes of HCl [1] and less readily hydrolysed.

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.54 cm, No bullets or numbering

Formatted: Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 5 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Font: 11 pt

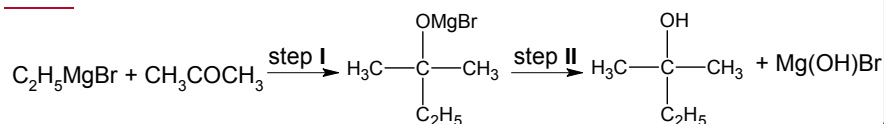
Formatted: Font: (Asian) Calibri, 11 pt, Complex Script Font: 11 pt, English (Singapore)

Formatted: Normal, Justified, Indent: Left: 0 cm, Don't add space between paragraphs of the same style

Formatted: Space After: 0 pt, Line spacing: single

Formatted: Font: 11 pt

(b) In 1911, the French chemist F.A.V. Grignard reacted small pieces of magnesium with a warm solution of bromoethane in a dry, non-polar solvent and obtained a solution containing ethylmagnesium bromide, $\text{C}_2\text{H}_5\text{MgBr}$. Many Grignard reagents, with different alkyl or aryl groups, have now been prepared and are widely used in organic syntheses. A typical example of the use of a Grignard reagent is the two-step reaction of $\text{C}_2\text{H}_5\text{MgBr}$ with propanone, CH_3COCH_3 , to form 2-methylbutan-2-ol.



Formatted: Font: 11 pt, Complex Script Font: 11 pt

Suggest the type of reaction which occurs in step II. [1]

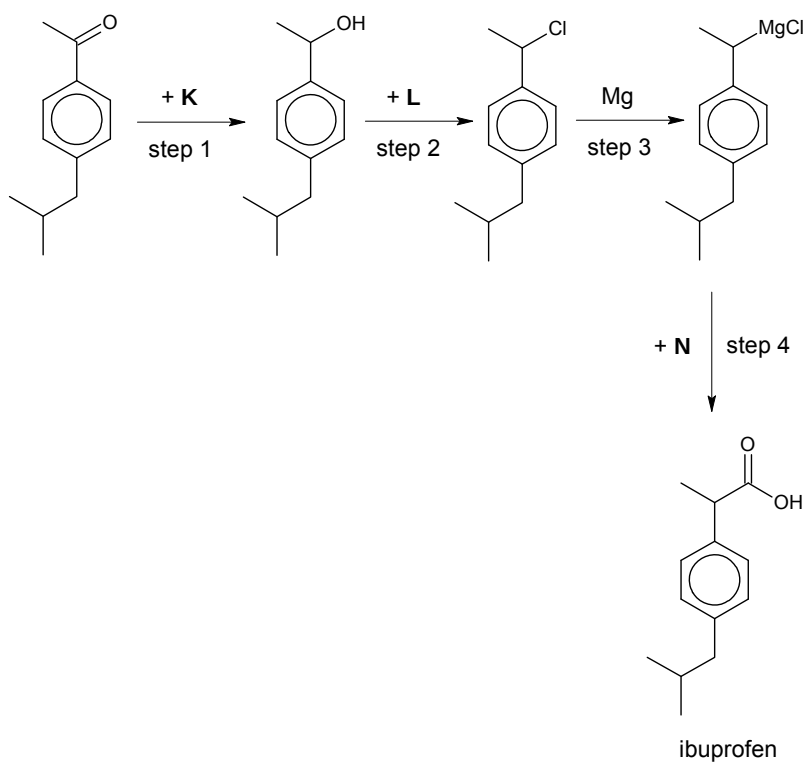
Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Hydrolysis [1]

Formatted: Font: Arial, 11 pt

(c) The following scheme shows the synthesis of ibuprofen which is an alternative medication to aspirin. In step 4, the Grignard reagent readily converts into a carboxylic acid.



(i) Suggest the identity of the reagent K in step 1.

NaBH₄ [1]

(ii) Suggest the identity of the reagent L in step 2.

PCl₅ or SOCl₂ [1]

(iii) Suggest the identity of the reagent N in step 4.

CO₂ [1]

(d) Suggest a simple chemical test that could be used to distinguish between aspirin and ibuprofen. You should state what you would observe for each compound.

1. Add H₂SO₄(aq), heat [1] – test
2. Distill

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

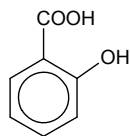
Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

12



Product obtained from aspirin: CH_3COOH .

3. Add $\text{Br}_2(\text{aq})$.

Orange Br_2 decolourises with aspirin; Orange Br_2 remains for ibuprofen. [1]

[Total: 16]

Copper(I) sulfate, Cu_2SO_4 , can be made from copper(I) oxide under non-aqueous conditions. On adding this salt to water, it immediately undergoes a disproportionation reaction.

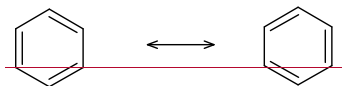
(i) Suggest, with a reason, the colour of copper(I) sulfate. [2]

(ii) Using suitable data from the Data Booklet, explain why the disproportionation reaction occurs, and write an equation for it. [3]

(b) Palladium(II) salts can form square planar complexes. Successive addition of ammonia and hydrogen chloride to an aqueous palladium(II) salt produces, under different conditions, three compounds with empirical formula $\text{PdN}_2\text{H}_6\text{Cl}_2$. Two of these, A and B, are non-ionic, with $M_r = 241$. A has a dipole moment, whereas B has none. The third compound, C, is ionic, having $M_r = 422$, and contains palladium in both its cation and anion.

For each A, B and C, deduce a structure that fits the above data, explaining your reasons fully. [6]

(c) Benzene ring is often represented as a structure that has a ring within the hexagon. Alternatively, chemists have also represented the structure of benzene in the following forms, known as resonance structures.

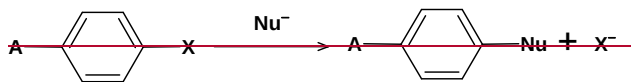


Benzene (two resonance forms)

The resonance relationship is indicated by the double headed arrow between them. The only difference between resonance forms is the placement of the pi electrons and non-bonding electrons.

Most aromatic compounds undergo electrophilic substitution. However aryl halides undergo a limited number of substitution reactions with strong nucleophiles.

An example of a reaction is as follows:



where A is an electron withdrawing group and X is a halogen

The mechanism of this reaction has two steps:

- addition of the nucleophile
- elimination of the halogen leaving group

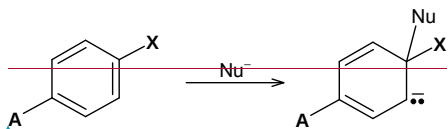
Formatted: Font: 11 pt

Formatted: Font: 11 pt

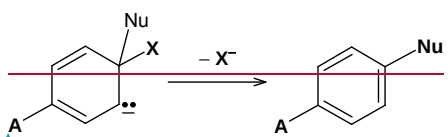
Formatted: Font: 11 pt

Formatted: Indent: First line: 0 cm

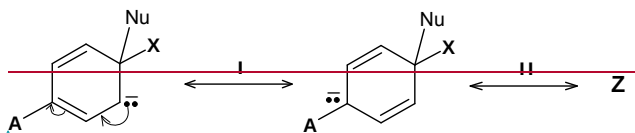
3 Step 1 involves the addition of the nucleophile (Nu^-). The Nu^- attacks the carbon atom bonded to a halogen, causing the pi bond to break. A resonance stabilised carbanion with a new $\text{C}=\text{Nu}$ bond is formed. The aromatic ring is destroyed in this step.



Step 2 involves the loss of the halogen X , reforming the aromatic ring.

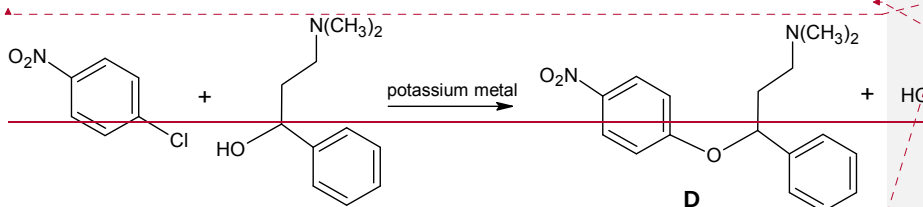


Two other resonance structures of the intermediate in **Step 1** are shown below:



(i) Copy the above diagram and draw the resonance structure, **Z**. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons in forming **Z**. [2]

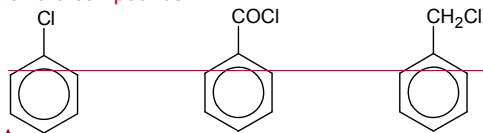
The reaction below shows the synthesis of compound, **D**.



(ii) Suggest the role of potassium metal in the reaction. [1]

(iii) Use the information given above to draw out the full mechanism for the reaction that forms **D**, labelling the slow and fast steps. In your answer, showing any relevant charges, lone pair of electrons and movement of electrons. [3]

(d) Describe and explain the relative ease of hydrolysis of the following three chloro compounds. [3]



[Total: 20]

Formatted: Indent: First line: 0 cm

2 Sulfuric acid, H_2SO_4 , can behave as an acid, an oxidising agent or as a dehydrating agent in various reactions.

(a) Draw a diagram to illustrate the shape of pure sulfuric acid and indicate the bond angle about the sulfur atom. [2]

(b) The Contact Process is used for the manufacture of sulfuric acid. One of the reactions that takes place is the following reversible reaction:



Sulfur dioxide and oxygen in a 2:1 molar ratio at a total initial pressure of 3 atm is passed over a catalyst in a fixed volume vessel at 400°C . When equilibrium is established, the percentage of sulfur trioxide in the mixture of gases is found to be 30%.

(i) Write an expression for the equilibrium constant, K_p , of the reaction. [1]

(ii) Calculate the value of K_p at 400°C , stating its units. [3]

(iii) How would the percentage conversion of SO_2 into SO_3 be affected when the pressure is raised? Explain. [2]

(c) Dilute sulfuric acid takes part in typical acid base reactions and it can be used to distinguish the following solids: MgO , BaO and SiO_2 .

State the observations, if any, to indicate the differences in their reaction when water is added to each solid followed by dilute sulfuric acid. [4]

(d) Sulfur dioxide is a major pollutant from sulfuric acid plants. The SO_2 emitted into the atmosphere is oxidised in the air, which then reacts with water to form sulfuric acid, hence causing acid rain:



Using the data below and data from (b), construct an energy cycle to calculate

(i) the enthalpy change of formation of $\text{SO}_2(\text{g})$, and hence

(ii) the enthalpy change of reaction, ΔH_1 , for the above reaction.

Enthalpy change of formation of $\text{H}_2\text{O}(\text{l})$	= -286 kJ mol^{-1}
Enthalpy change of formation of $\text{H}_2\text{SO}_4(\text{l})$	= -811 kJ mol^{-1}
Enthalpy change of formation of $\text{SO}_3(\text{g})$	= -493 kJ mol^{-1}
[4]	

Formatted: Justified, Indent: Left: 1.27 cm, Right: 0 cm

Formatted: Indent: First line: 0 cm, Right: 0 cm

Formatted: Indent: First line: 0 cm, Automatically adjust right indent when grid is defined, Allow hanging punctuation, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers, Font Alignment: Auto

Formatted: Indent: First line: 0 cm

Formatted: Font: Arial

Formatted: Indent: Left: 1.27 cm

Formatted: Indent: Left: 1.27 cm, No bullets or numbering

Formatted: Indent: Left: 1.27 cm

Formatted: Indent: Left: 1.27 cm, No bullets or numbering

Formatted: Indent: Left: 1.27 cm

Formatted: Indent: Left: 1.27 cm, No bullets or numbering

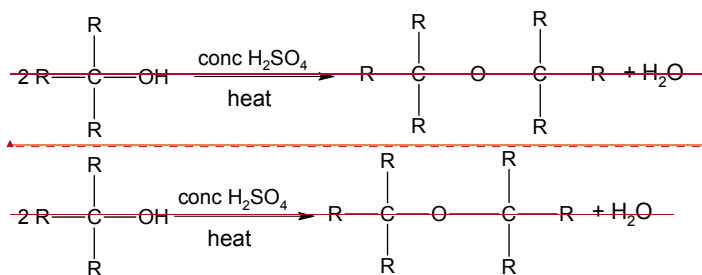
Formatted: Indent: First line: 0 cm, Right: 0 cm

Formatted: Indent: Left: 1.27 cm, First line: 0 cm, Automatically adjust right indent when grid is defined, Allow hanging punctuation, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers, Font Alignment: Auto

Formatted: Indent: Left: 1.27 cm, Right: 0 cm

Formatted: Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Allow hanging punctuation, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers, Font Alignment: Auto

(e) — Alcohols react with concentrated sulfuric acid at high temperatures to form alkenes. A common side reaction that can happen is the formation of ethers, which is also catalysed by concentrated sulfuric acid.



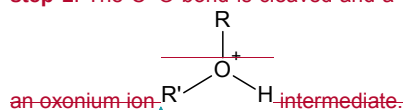
The mechanism occurs via 3 steps:

Step 1:-

An acid base reaction in which H^+ from H_2SO_4 protonates the oxygen atom in alcohol. This step is very fast and reversible.

Step 2:-

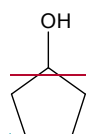
A second alcohol molecule functions as the nucleophile and attacks the product from step 1. The C-O bond is cleaved and a water molecule leaves the molecule. This creates



Step 3:-

Another acid base reaction in which the proton in the oxonium ion is removed by a suitable base (in this case a water molecule) to give the ether product. This step is very fast and reversible.

(i) — Draw the ether formed when cyclopentanol undergoes the above reaction.



cyclopentanol
[1]

(ii) — Draw out the full mechanism for the reaction between two cyclopentanol molecules to form an ether. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons. [3]

— [Total: 20]

Formatted: Font: Arial, 11 pt, Complex Script Font: 11 pt

Field Code Changed

Formatted: Add space between paragraphs of the same style

Formatted: Justified, Add space between paragraphs of the same style

Formatted: Add space between paragraphs of the same style

Formatted: Space Before: 0 pt, After: 0 pt, Add space between paragraphs of the same style

Field Code Changed

Formatted: Add space between paragraphs of the same style

Formatted: Indent: Left: 1.27 cm, Add space between paragraphs of the same style

Formatted: Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Space After: 0 pt, Add space between paragraphs of the same style, No bullets or numbering, Allow hanging punctuation, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers, Font Alignment: Auto

Formatted: Justified, Indent: Left: 1.27 cm, Add space between paragraphs of the same style

Field Code Changed

Formatted: Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Add space between paragraphs of the same style, No bullets or numbering, Allow hanging punctuation, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers, Font

Formatted: Indent: First line: 0 cm, Right: 0 cm

Formatted: Justified, Indent: Left: 1.27 cm, Right: 0 cm

Formatted: Font: (Default) Arial, Complex Script Font: Arial

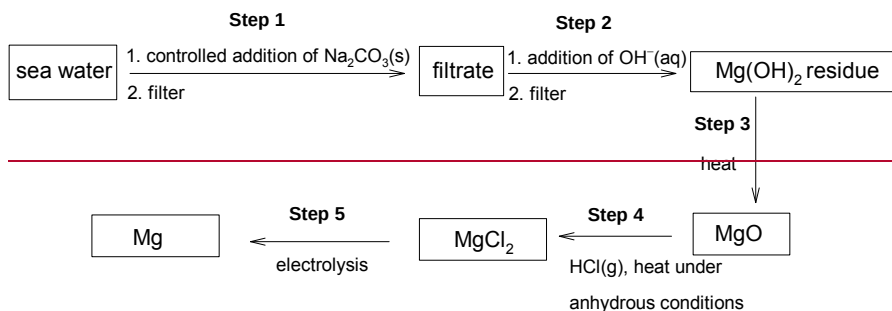
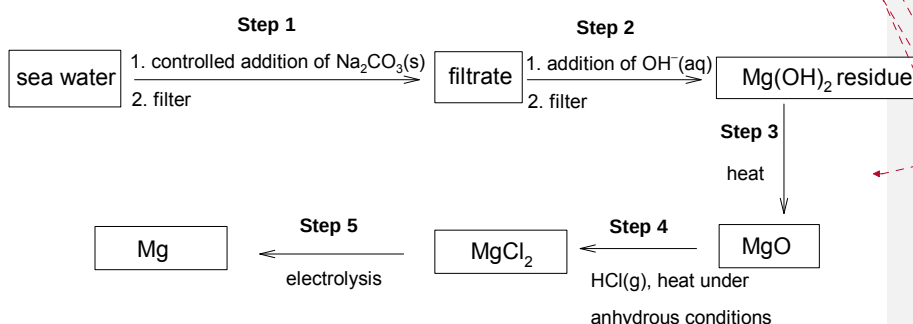
Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.27 cm,
Space After: 0 pt

- 3 Magnesium is present as dissolved magnesium ions in sea water and is the only metal directly extracted from sea water. There is enough magnesium dissolved in the Earth's oceans to supply all of our magnesium needs for the next 1000 years.

Garlic contains many amino acids, minerals and enzymes. Garlic also contains at least 33 sulfur compounds like alliin, allicin and ajoene. The sulfur compounds are responsible for both garlic's pungent odour and many of its medical effects.

- (a) Apart from magnesium ions, the two other most abundant cations found in sea water are sodium and calcium ions.

Magnesium can be extracted from sea water by the following steps:



Concentration of common ions in sea water:

ion	concentration / mol dm ⁻³
magnesium	0.056
calcium	0.010
sodium	0.457
chloride	0.535

The numerical values of solubility products are given below:

compound	value of solubility product
magnesium carbonate	1.00×10^{-5}
calcium carbonate	8.70×10^{-9}

Formatted: Font: (Default) Arial, Complex Script Font: Arial, English (Singapore), Highlight

Formatted: English (United States)

Formatted: Font: (Default) Arial, Complex Script Font: Arial

Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.27 cm, Space After: 0 pt, No bullets or numbering, Widow/Orphan control

Formatted: Font: (Default) Arial, Complex Script Font: Arial, Highlight

Formatted: Indent: First line: 0.63 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt

Formatted: Justified, Indent: Left: 0.63 cm, First line: 0.63 cm, Space After: 0 pt, Line spacing: single

Formatted: Font: Bold

Formatted: Font: Symbol, Bold

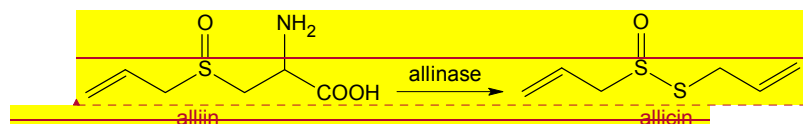
Formatted: Font: Bold

Formatted: Indent: First line: 1.27 cm

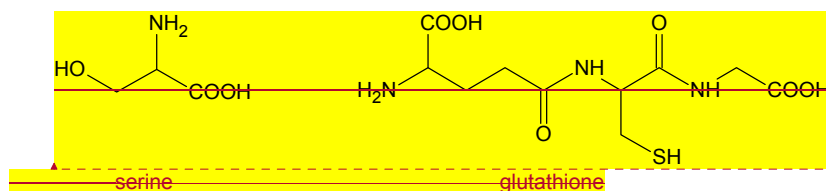
Formatted: Font: Bold

magnesium hydroxide	5.61×10^{-12}
calcium hydroxide	5.50×10^{-6}

When raw garlic is chopped or crushed, the enzyme allinase converts alliin into allicin, which is responsible for the odour of fresh garlic.



Allicin can also be biosynthesised by using serine and glutathione.



- (i) Explain why raw chopped garlic has a stronger odour than when it is cooked. [1]
 (i) Explain why the addition of sodium carbonate ions in step 1 has to be controlled. [1]

If too much carbonate ions was added, both MgCO_3 and CaCO_3 will be precipitated.

- (ii) Hence, state the cations present in the filtrate after step 1 is carried out. [1]

Na^+ and Mg^{2+}

- (iii) What is the maximum mass of solid sodium carbonate that can be added to 1 dm^3 of sea water in step 1? [2]

The $[\text{CO}_3^{2-}]$ must be controlled such that only calcium carbonate precipitates out, leaving MgCO_3 in the solution.

$$K_{\text{sp}} = [\text{Mg}^{2+}][\text{CO}_3^{2-}]$$

$$\text{maximum}[\text{CO}_3^{2-}] = \frac{K_{\text{sp}}}{[\text{Mg}^{2+}]} = \frac{1.0 \times 10^{-5}}{0.056} = 1.785 \times 10^{-4} \text{ [1]}$$

$$m_{\text{CaCO}_3} = 1.785 \times 10^{-4} \times 100.1 = 0.0179 \text{ g [1]}$$

- (iv) Use the data provided to explain the following:

- Solid sodium carbonate was added to sea water (under controlled conditions) before the hydroxide ions.
- The reverse order (i.e. adding hydroxide ions before sodium carbonate) is not preferred over the extraction of magnesium, (under controlled conditions)

sodium

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Left, Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Space After: 8 pt, Don't add space between paragraphs of the same style, No bullets or numbering, Widow/Orphan control, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

Formatted: Font: 11 pt

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

Formatted: Left, Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Space After: 8 pt, Don't add space between paragraphs of the same style, No bullets or numbering, Widow/Orphan control, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

Formatted: Indent: First line: 1.23 cm

Formatted: Left, Indent: Left: 1.27 cm, Automatically adjust right indent when grid is defined, Space After: 8 pt, Don't add space between paragraphs of the same style, No bullets or numbering, Widow/Orphan control, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

Formatted: Font: Bold

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Font color: Auto, Complex Script Font: 11 pt

Formatted: Indent: Left: 2.5 cm

Formatted: Indent: First line: 1.23 cm

Formatted: Font: 11 pt, Font color: Auto, Complex Script Font: 11 pt

Formatted: Automatically adjust right indent when grid is defined, Space After: 8 pt, Don't add space between paragraphs of the same style, No bullets or numbering, Widow/Orphan control, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt

Formatted: Font: (Default) Arial, Complex Script Font: Arial, English (Singapore)

Formatted: Font: (Default) Arial, Bold, Complex Script Font: Arial

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt, English (Singapore)

Formatted: Left, Indent: Hanging: 0.04 cm, Space After: 0 pt

Formatted: Left, Space After: 0 pt, No bullets or numbering

Formatted: Right, Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

Comparing the K_{sp} of the 2 carbonates, CaCO_3 is less soluble and will be precipitated out first, leaving the Mg^{2+} ions in solution, and can be further purified via other steps. [1]

If hydroxide ions was added before carbonate ions, $\text{Mg}(\text{OH})_2$ having a smaller K_{sp} value than $\text{Ca}(\text{OH})_2$ will be less soluble and precipitate out first, together with other solid impurities in sea water, leading to an impure product. [1]

(v) Calculate the minimum pH of the hydroxide solution required for precipitation of magnesium hydroxide in step 2 if an equal volume of hydroxide ions was added to the filtrate. Give your answer to 2 decimal places. [2]

Assume $[\text{Mg}^{2+}] = 0.056 \text{ mol dm}^{-3}$ (from table)

After adding equal volume of hydroxide ions, $[\text{Mg}^{2+}]_{\text{new}} = \frac{0.056}{2} = 0.028 \text{ mol dm}^{-3}$

$$K_{sp} = [\text{Mg}^{2+}][\text{OH}^-]^2$$

$$[\text{OH}^-] = \sqrt{\frac{K_{sp}}{[\text{Mg}^{2+}]}} = \sqrt{\frac{5.61 \times 10^{-12}}{0.028}} = 1.415 \times 10^{-5} \quad [1]$$

$$[\text{OH}^-]_{\text{required}} = 1.415 \times 10^{-5} \times 2$$

$$\text{pOH} = -\log(1.415 \times 10^{-5} \times 2) = 4.55$$

$$\text{pH} = 9.45 \quad [1] \quad 2\text{dp ans}$$

(b) (i) Write the equations that occur during the electrolysis of magnesium chloride in Step 5. State clearly the reactions that occur at the cathode and the anode, and include state symbols. [2]

Cathode: $\text{Mg}^{2+}(\text{l}) + 2\text{e}^- \rightarrow \text{Mg}(\text{l})$ [1]

Anode: $2\text{Cl}^-(\text{l}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$ [1]

(ii) In a factory, a current of 95 kA was passed through a suitable setup for 24 hours. Assuming that the procedure is 90% efficient, calculate the mass of Mg that can be produced. [2]

$$Q = It = n z F$$

$$\frac{95 \times 1000 \times 24 \times 60 \times 60}{100} \times 90 = n \times 2 \times 96500$$

$$n = 38.28 \times 10^4 \quad [1]$$

$$m = 38.28 \times 10^4 \times 24.3 = 930,000 \text{ g} = 930 \text{ kg} \quad [1]$$

(iii) Give a reason why electrolysis of magnesium chloride is preferred to that of magnesium oxide in this industrial process. [1]

MgCl_2 has a lower melting point than MgO , hence the electrolysis can be carried out at a lower temperature, saving operation costs. [1]

Formatted: Indent: Left: 2.54 cm

Formatted: Right, Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

Formatted: Font: Bold, Font color: Auto

Formatted: Indent: First line: 1.27 cm

Formatted: Indent: First line: 1.23 cm

Formatted: Default, Indent: Left: 1.27 cm, Hanging: 1.27 cm, Automatically adjust right indent when grid is defined, Space After: 0 pt, Line spacing: single, No bullets or numbering, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

Formatted: Font: Not Bold

Formatted: Indent: Left: 1.87 cm, First line: 0.63 cm

Formatted: Indent: Left: 1.27 cm, Hanging: 1.27 cm, Space After: 0 pt, No bullets or numbering

Formatted: Space After: 0 pt, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

Formatted: Indent: Left: 1.27 cm

Formatted: Space After: 0 pt, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Default, Indent: Left: 1.27 cm, Hanging: 1.27 cm, Automatically adjust right indent when grid is defined, Line spacing: single, No bullets or numbering, Adjust space between Latin and Asian text, Adjust space between Asian text and numbers

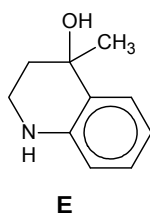
Formatted: Indent: Left: 2.54 cm

Formatted: Indent: Left: 0 cm, Hanging: 1.27 cm, Space After: 0 pt, No bullets or numbering

(c) A is an organic compound. When 0.4678 g of an organic compound A was evaporated in a syringe, the volume of the vapour produced after correction to s.t.p was 60 cm³. On heating with aqueous sodium hydroxide, A gives a compound that dissolves in water.

A reacts with aluminium oxide to give two products B and C. Both B and C react with HBr to give the same product D. D exhibits enantiomerism and exists as a pair of enantiomers.

A gives E when reacted with lithium aluminium hydride in dry ether.



(i) Prove that the molar mass of A is 177 g mol⁻¹.

[1]

From Data Booklet, at s.t.p, $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ at 10⁵ Pa and 273 K

Either

$$10^5 \times 60 \times 10^{-6} = \frac{0.468}{M} \times 8.31 \times 273$$

$$M = 176.9 = 177 \text{ g mol}^{-1} \text{ [1]}$$

Or

$$n(A) = 60 / 22700 = 2.643 \times 10^{-7} \text{ mol}$$

$$M(A) = 0.468 / 2.643 \times 10^{-7} = 177.06 \approx 177 \text{ g mol}^{-1}$$

Comments

- Generally well-done.
- Common mistake include:
Using incorrect V_m such as $V_m = 24 \text{ dm}^3$ or $V_m = 22.4 \text{ dm}^3$
- A small number of students did not use the numerical data of 0.468 g and 60 cm³ to prove the molar mass of A. These students used the following approach, which is NOT acceptable for this question.
1 mol of A undergoes reduction to give 1 mol of E.
molar mass of E = 163.0 g mol⁻¹
A has amide.
molar mass of A = 163.0 - 2(1.0) + 16.0 = 177.0 g mol⁻¹
Note that you are expected to use the numerical data of 0.468 g and 60 cm³ to prove the molar mass of A.

(ii) Hence, deduce the structural formulae of all the above structures, and explain the

Formatted: Indent: Left: 1.27 cm, Space After: 0 pt

Formatted: Indent: First line: 0.63 cm, Space After: 0 pt

Formatted: Centered, Indent: First line: 0.63 cm

Formatted: Space After: 0 pt, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 1 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Font color: Blue

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

Formatted: Lowered by 20 pt

Formatted: Indent: First line: 0.63 cm

Formatted: Superscript

Formatted: Font: Italic, Complex Script Font: Italic

Formatted: Font: Arial

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Bulleted + Level: 3 + Aligned at: 3.17 cm + Indent at: 3.81 cm

Formatted: Indent: Left: 3.17 cm, First line: 0.63 cm, Space After: 0 pt

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Bulleted + Level: 3 + Aligned at: 3.17 cm + Indent at: 3.81 cm

Formatted: Font: 10 pt, Complex Script Font: 10 pt

Formatted: Indent: Left: 3.17 cm, Space After: 0 pt

Formatted: Font: 10 pt, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Bulleted + Level: 3 + Aligned at: 3.17 cm + Indent at: 3.81 cm

Formatted: Indent: Left: 3.17 cm, First line: 0.63 cm, Space After: 0 pt

Formatted: Indent: Left: 3.17 cm, Space After: 0 pt

Formatted: Font: 10 pt, Complex Script Font: 10 pt

Formatted: Indent: Left: 3.17 cm, First line: 0.63 cm, Space After: 0 pt

Formatted: Font: 10 pt, Bold, Complex Script Font: 10 pt, Bold

Formatted: Indent: Left: 3.17 cm, Space After: 0 pt

Formatted: Font: 10 pt, Complex Script Font: 10 pt

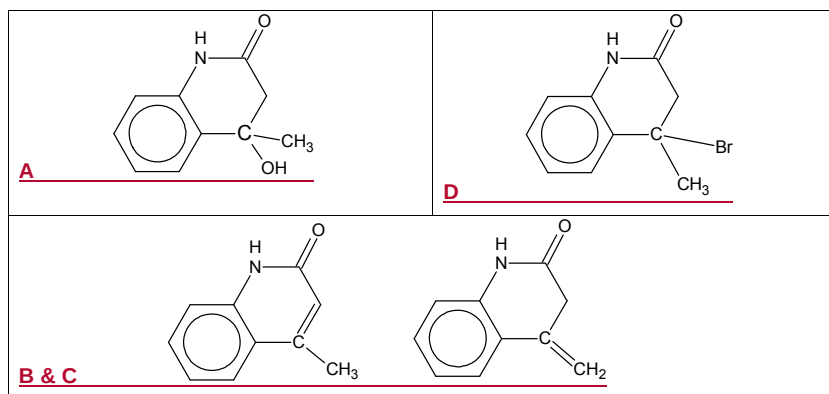
Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt

Formatted: Space After: 0 pt

chemistry involved.

[6]



1 mk for each structure [4]

1/2 mk for each point (max of 2 mks):

- From the molar mass of 177 g mol^{-1} , the molecular formula is likely to be $\text{C}_{10}\text{H}_{11}\text{O}_2\text{N}$.
- A** undergoes basic hydrolysis with NaOH (aq) to give COO^- ($\text{COOH} + \text{NaOH}$) and NH_2 . COO^- is an ionic salt that is soluble in water by forming ion-dipole interactions with water. $\Rightarrow$ **A** contains an amide group to give an amine and carboxylic acid when hydrolysed.
Some students wrote that the compounds from the hydrolysis of **A** dissolve in water to via formation of hydrogen bonds with water. It may seem correct, but note that **A** contains an amide in a cyclic structure, hence after hydrolysis, there will be only one single product.
- A** undergoes elimination with Al_2O_3 to give alkenes **B** and **C** $\Rightarrow$ **A** contains an alcohol group
- The amide group in **A** undergoes reduction with LiAlH_4 to give an amine in **E**
- Alkenes **B** and **C** undergo electrophilic addition with HBr to give halogenoalkane/alkyl halide/bromoalkane **D**.
- D** contains a chiral carbon and hence exists as a pair of enantiomers.

Comments

Out of the above 6 statements, the following key points are expected to gain credit.

- A** undergoes (basic) hydrolysis with NaOH $\Rightarrow$ **A** contains an amide group
- A** undergoes elimination with Al_2O_3 to give alkenes **B** and **C** $\Rightarrow$ **A** contains an alcohol group
- The amide group in **A** undergoes reduction with LiAlH_4 to give an amine in **E**
- Alkenes **B** and **C** undergo electrophilic addition with HBr to give halogenoalkane/alkyl halide/bromoalkane **D**.

OR

- D** contains a chiral carbon and hence exists as a pair of enantiomers.

4 bullet pts correct – 2 marks

2 bullet pts correct (minimum) – 1 mark

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

Formatted: Font: Not Bold, Complex Script Font: Bold

Formatted: Indent: Left: 1.9 cm, No bullets or numbering

Formatted: Complex Script Font: Bold

Formatted: Indent: Left: 2.54 cm, No bullets or numbering

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Bulleted + Level: 3 + Aligned at: 3.17 cm + Indent at: 3.81 cm, Widow/Orphan control

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt, Highlight

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt, Highlight

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt, Highlight

Formatted: Highlight

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Font: 10 pt, Bold, Underline, Font color: Auto, Complex Script Font: 10 pt, Bold

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Font: Not Bold

Formatted: Indent: Left: 2.54 cm

Formatted: Indent: Left: 3.17 cm, No bullets or numbering

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm

Formatted: Font: 10 pt, Bold, Underline, Font color: Auto, Complex Script Font: 10 pt, Bold

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Font: 10 pt, Bold, Underline, Font color: Auto, Complex Script Font: 10 pt, Bold

Formatted: Font: 10 pt, Underline, Font color: Auto, Complex Script Font: 10 pt, Bold

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: List Paragraph, Left, No bullets or numbering

Formatted: Font: Bold, Complex Script Font: Bold

Formatted: Indent: Left: 3.17 cm, No bullets or numbering

Formatted: Font: 10 pt, Underline, Font color: Auto, Complex Script Font: 10 pt

Formatted: Font: 10 pt, Bold, Underline, Font color: Auto, Complex Script Font: 10 pt

Formatted: Font: 10 pt, Font color: Auto, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm

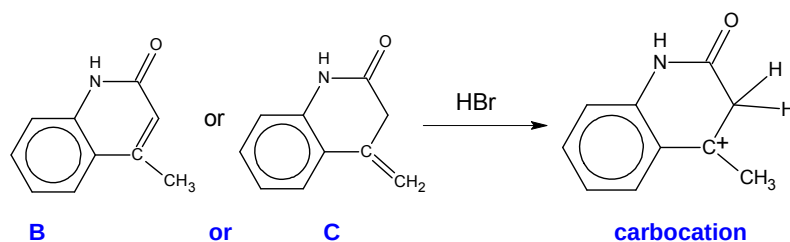
Formatted: Font: 10 pt, Complex Script Font: 10 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, No bullets or numbering

- (iii) State the type of isomerism exhibited by **B** and **C**. Explain why **B** and **C** both give the same product **D** when reacted with HBr. [2]

Constitutional isomerism. [1]

When alkenes **B** and **C** undergo electrophilic addition with HBr, both form the same carbocation. The carbocation will be attacked by Br⁻, which leads to the formation of **D**.



[1] for explanation & structure of carbocation

- (ii) When one molecule of serine reacts with one molecule of glutathione, it is possible to form two esters with different structural formulae. Draw the structural formula of **each** of these esters. [2]

- (iii) Draw the structural formulae of the products when glutathione is hydrolysed. [3]

Alliin has pK_a values of 1.84 and 8.45.

- (iv) Make use of these pK_a values to suggest the major species present in solutions of alliin with the following pH values. [3]

- pH 1
- pH 7
- pH 11

- v) Calculate the pH of 0.10 mol dm⁻³ solution of alliin. [1]

- (vi) With reference to the pK_a values, identify the major species formed when 10 cm³ of 0.10 mol dm⁻³ NaOH is added to 10 cm³ of 0.10 mol dm⁻³ protonated alliin. Hence, deduce whether the solution is acidic, neutral or alkaline. [2]

- (vii) Sketch the pH volume added curve you would expect to obtain when 30 cm³ of 0.10 mol dm⁻³ NaOH is added to 10 cm³ of 0.10 mol dm⁻³ protonated alliin. Briefly describe how you have calculated the various key points on the curve. [4]

Formatted: Space After: 0 pt, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 1 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Font color: Blue

Formatted: Indent: Left: 2.54 cm

Formatted: Indent: First line: 0.63 cm

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

Formatted: Not Highlight

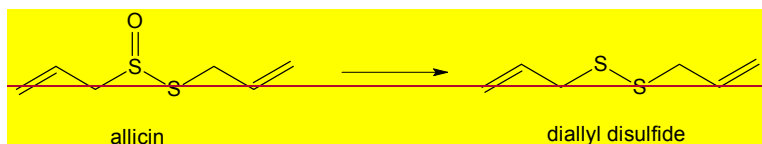
Formatted: Right

Formatted: Right

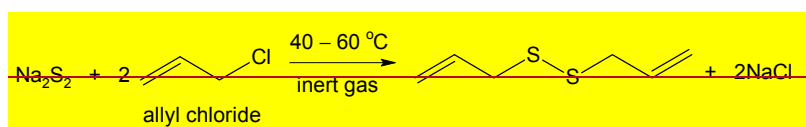
Formatted: Not Highlight

Formatted: Highlight

- (b) Diallyl disulfide is one of the principal components of the distilled oil of garlic. It is a yellowish liquid which is insoluble in water and has a strong garlic odour. It is produced during the decomposition of allicin.



Diallyl disulfide can be produced industrially from sodium disulfide and allyl chloride at temperatures of $40 - 60\text{ }^{\circ}\text{C}$ in an inert atmosphere.



- (i) Give the IUPAC name of allyl chloride. [1]
- (ii) Explain, in thermodynamic terms, suggest why diallyl disulfide is insoluble in water. [2]
- (iii) State the type of reaction when diallyl disulfide is converted back to allicin. [1]

[Total: 20]

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Section B

Answer one question from this section.

- 4 Cycloalkanes are a homologous series of cyclic saturated hydrocarbons with the general formula C_nH_{2n} while n-alkanes are a homologous series of straight-chain saturated hydrocarbons with the general formula C_nH_{2n+2} .

n-alkanes	Boiling point / °C	Enthalpy change of combustion / kcal mol ⁻¹	cycloalkanes	Boiling point / °C	Enthalpy change of combustion / kcal mol ⁻¹
ethane	-89	-373.0	=	=	=
propane	-42	-530.4	cyclopropane	-33	-499.8
butane	-1	-687.8	cyclobutane	12	-656.0
pentane	36	-845.2	cyclopentane	49	-793.5
hexane	69	-1002.6	cyclohexane	81	-944.6
heptane	98	-1160.0	cycloheptane	119	-1108.3

(a) Explain the term "homologous series".

[1]

Group of organic compounds with

- Same functional group
- Differ by a CH_2 group

(b) Explain the increase in magnitudes of both boiling point and enthalpy change of combustion from ethane to heptane.

[3]

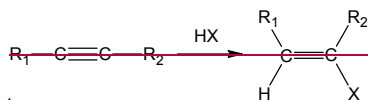
- Alkanes have simple molecular structure with weak instantaneous dipole-induced dipole (id-id) forces between molecules
- From ethane to heptane, the electron cloud gets bigger and more easily polarised, hence id-id forces become stronger.
- More energy needed to break the id-id forces, hence boiling point increases
- From ethane to heptane, the molecule increases by a CH_2 group
- This results in more C – C bonds and C – H bonds to break and more C=O and O – H bonds to form, hence more heat given off during combustion.

✓✓✓

Alkynes are organic molecules which contain carbon-carbon triple bonds and are part of the homologous series with formula of C_nH_{2n-2} i.e.

where R_1 and R_2 = H or alkyl or aryl groups

Alkynes exhibit similar chemical properties to alkenes.

e.g. addition reactions with electrophiles i.e. X_2 or HX to form alkenes

Formatted: Font: 11 pt

Formatted: Centered

Formatted: Font: 10 pt

Formatted: Indent: Left: 0 cm, Hanging: 1.25 cm, Tab stops: 0.75 cm, Left

Formatted: Font: 11 pt

Formatted: Font: 11 pt, Bold, Complex Script Font: 11 pt, Bold, Italic, English (United Kingdom), Highlight

Formatted: Font: 11 pt, Complex Script Font: 11 pt, Not Highlight

Formatted: Normal, No bullets or numbering

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt

Formatted: Indent: Left: 1.39 cm, Hanging: 0.63 cm, Space After: 0 pt, Line spacing: single

Formatted: Normal, No bullets or numbering

Formatted: Font: 11 pt, Bold

Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.27 cm, No bullets or numbering

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

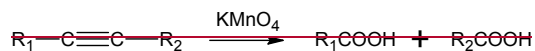
Formatted: Font: 11 pt

Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.27 cm, No bullets or numbering

Field Code Changed

Formatted: Font: 11 pt, Complex Script Font: 11 pt

e.g. oxidation by hot concentrated KMnO_4 to form mixture of carboxylic acids



However, unlike alkenes, terminal alkynes are able to react with strong bases like sodium amide.



- (a) Ethyne, C_2H_2 , is heated with excess sodium bromide and concentrated sulfuric acid to produce a dihalide, $\text{C}_2\text{H}_4\text{Br}_2$. The overall reaction may be considered to take place in two stages, the first between inorganic reagents only and the second involving the organic reagent.

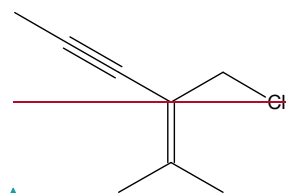
(i) Write an equation for the first stage. [1]

(ii) Suggest a structure for the dihalide formed. [1]

(iii) When the concentrated sulfuric acid is added to the reaction mixture, cooling is necessary to prevent the formation of inorganic by-products.

Write an equation to explain the formation of these inorganic by-products. [1]

- (b) Compound A, is an enyne chloride (i.e. compounds that contains chloro, alkyne and alkene functional groups).



Compound A

One mole of compound A reacts with two moles of Br_2 to produce a mixture of 4 stereoisomers. Draw structures of the stereoisomers formed. [3]

- (c) Compound B, which is an isomer of Compound A and also an enyne chloride, is treated with sodium amide, NaNH_2 followed by heating under reflux to form compound C, C_8H_{10} . Compound C reacts with hot concentrated KMnO_4 to produce butane-1,4-dioic acid only. Suggest why combustion tends to be incomplete as the alkane increases in molecular mass. [1]

As the alkane increases in molecular mass, the alkane becomes liquid and liquid phase reactions are slower than gas phase reactions. Hence combustion tends to be incomplete.

Cyclopropane is a colourless gas with a "petroleum-like" odour. Unlike its straight-chain counterpart, it is considered to be highly strained and unstable. The instability of cyclic alkanes can be measured by calculating its "ring strain energy" using the formula below:

Ring strain energy =

$$\text{number of carbon atoms in cyclic alkane, } A \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane, } A \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right)$$

$$\text{number of carbon atoms in cyclic alkane, } A \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane, } A \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right)$$

$$\begin{aligned} & \text{number of carbon atoms in cyclic alkane, } A \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane, } A \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right) \\ & \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \\ & \text{number of carbon atoms in cyclic alkane, } A \\ & \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane, } A \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right) \end{aligned}$$

$$\begin{aligned} & \text{number of carbon atoms in cyclic alkane, } A \\ & \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane, } A \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right) \\ & \text{number of carbon atoms in cyclic alkane} \times \left(\left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in the} \\ \text{cyclic alkane} \end{array} \right| - \left| \begin{array}{c} \text{enthalpy change of combustion} \\ \text{of a CH}_2 \text{ group in an} \\ \text{unstrained n-alkane} \end{array} \right| \right) \end{aligned}$$

(d) Given that the ΔH_f of CO_2 is $-94.05 \text{ kcal mol}^{-1}$ and ΔH_f of water is $-68.3 \text{ kcal mol}^{-1}$, write an equation showing the formation of cyclopropane, and hence calculate the enthalpy change of formation of cyclopropane. calculate the enthalpy change of formation of cyclopropane. [2+]



$$\Delta H_f = [3(-94.05) + 3(-68.3)] - (-499.8) = +12.75 \text{ kcal mol}^{-1}$$

(e) Using the formula above, prove that the ring strain energy in cyclopropane is $+27.6 \text{ kcal mol}^{-1}$. [2]

$$\begin{aligned} & \text{Enthalpy change of combustion of CH}_2 \text{ group in unstrained molecule} \\ & = -530.4 - (-373.0) = -157.4 \text{ kcal mol}^{-1} \text{ (can use other values to subtract)} \\ & \text{Enthalpy change of combustion of CH}_2 \text{ group in cyclopropane} \\ & = 1/3 (-499.8) = -166.6 \text{ kcal mol}^{-1} \\ & \text{Strain energy of cyclopropane} = 3 \times (166.6 - 157.4) = +27.6 \text{ kcal mol}^{-1} \end{aligned}$$

Formatted: Centered

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Centered

Formatted: English (United States), Not Highlight

Formatted: List Paragraph, Indent: Hanging: 1.27 cm, Space After: 0 pt, Add space between paragraphs of the same style, Line spacing: single, No bullets or numbering, Tab stops: 0.75 cm, Left + Not at 2.75 cm + 16.75 cm

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: List Paragraph, Indent: Hanging: 1.27 cm, Space After: 0 pt, Add space between paragraphs of the same style, Line spacing: single, No bullets or numbering, Tab stops: 0.75 cm, Left + Not at 2.75 cm + 16.75 cm

Formatted: Font: Bold

Formatted: No bullets or numbering

Formatted: Font: 11 pt

Formatted: Indent: First line: 1.27 cm

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

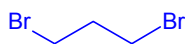
Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: No bullets or numbering

(f) Due to the presence of ring strain, cyclopropane undergoes an addition reaction with bromine in the absence of ultraviolet radiation.

(i) Suggest the skeletal structure of the molecule formed after reaction with Br₂. [1]



(ii) Hence, using VSEPR theory, explain why the presence of ring strain causes cyclopropane to undergo addition reactions. [1]

Cyclopropane has a C – C – C angle of 60°, which is smaller than the optimal angle of 109.5° in sp³ carbons. As a result, bond pairs are much closer to each other and experience greater repulsion, hence weakening the C – C bonds, which breaks easily during addition.

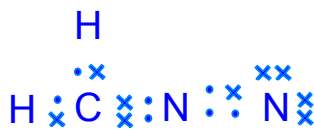
(g) Cyclopropane rings can be formed using a technique called “cyclopropanation”.

One such cyclopropanation technique involves the 2 mechanistic steps stated below:

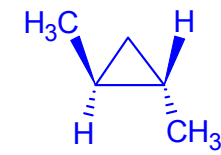
Step 1: Dissociation of diazomethane, CH₂N₂ to form methylene, CH₂, and N₂. Formation of methylene, CH₂, and N₂ from diazomethane, CH₂N₂.

Step 2: Addition of methylene, CH₂, to trans-but-2-ene to form the cyclic ring. The reaction leaves the stereochemistry of the molecule unchanged.

(i) It is observed that the diazomethane molecule is trigonal planar in shape. By considering the shape, draw a dot-and-cross diagram of diazomethane, CH₂N₂, clearly showing the type of bonds formed within the molecule. [1]



(ii) Draw the structure of the cyclic molecule formed in Step 2, showing the stereochemical arrangement clearly. State and explain if the molecule can rotate plane-polarised light. [2]



Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Hanging: 1.27 cm, Numbered + Level: 1 + Numbering Style: a, b, c, ... + Start at: 6 + Alignment: Left + Aligned at: 0.63 cm + Indent at: 1.27 cm, Tab stops: Not at 2.75 cm

Formatted: Indent: Hanging: 1.4 cm, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 1 + Alignment: Left + Aligned at: 1.38 cm + Indent at: 2.65 cm, Tab stops: Not at 16.75 cm

Formatted: Indent: Left: 2.65 cm, No bullets or numbering, Tab stops: Not at 16.75 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.5 cm

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: English (Singapore)

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Hanging: 1.27 cm, Numbered + Level: 1 + Numbering Style: a, b, c, ... + Start at: 6 + Alignment: Left + Aligned at: 0.63 cm + Indent at: 1.27 cm, Tab stops: Not at 2.75 cm

Formatted: Indent: First line: 0.52 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 0 cm, First line: 1.27 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 1.25 cm, Hanging: 1.44 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 1.25 cm, Hanging: 1.25 cm, Numbered + Level: 1 + Numbering Style: a, b, c, ... + Start at: 9 + Alignment: Left + Aligned at: 2.54 cm + Indent at: 3.17 cm, Tab stops: Not at 16.75 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 3.17 cm, No bullets or numbering, Tab stops: Not at 16.75 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

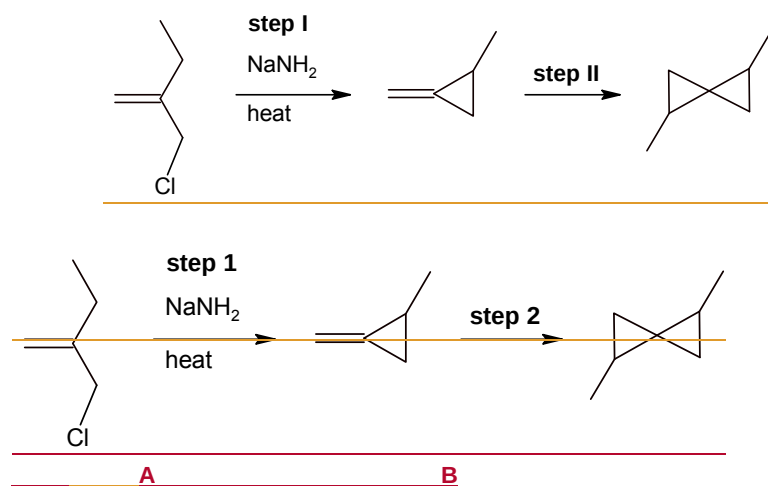
Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 0 cm

Yes, the molecule contains two chiral centres and does not have an internal plane of symmetry.

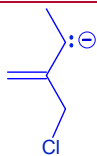
(h) Cyclopropane rings are a precursor for many types of fatty acids. The following shows part of the synthetic route for fatty acids.



(iii) Step 1 involves the reaction of molecule A with NaNH₂ to form NH₃ and a negatively-charged organic intermediate which eventually formed molecule B upon heating.

State the type of reactions that took place in step 1 and draw the organic intermediate that was formed.

- Molecule A reacted with NaNH₂ in an acid base reaction



- (Intramolecular) nucleophilic substitution

*✓✓

(iv) By considering the reactivity of the Cl atom, explain why molecule C cannot be used to replace molecule A in the synthesis above.

[12]

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.54 cm

Formatted: Indent: Left: 0 cm

Formatted: Font: 11 pt, Bold, Complex Script Font: 11 pt, Not Highlight

Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.27 cm, No bullets or numbering, Tab stops: 0.75 cm, Left

Formatted: Font: 11 pt, Complex Script Font: 11 pt, Not Highlight

Formatted: Font: 11 pt

Formatted: Font: 11 pt, Bold, Complex Script Font: 11 pt, Highlight

Formatted: Centered

Formatted: List Paragraph, Indent: Left: 1.27 cm, Hanging: 1.27 cm, Space After: 0 pt, Add space between paragraphs of the same style, Line spacing: single, No bullets or numbering, Tab stops: 0.75 cm, Left

Formatted: (Complex) Arabic (Saudi Arabia), Highlight

Formatted: Font: 11 pt

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Line spacing: single

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Indent: Left: 2 cm, No bullets or numbering

Formatted: Indent: Left: 0 cm

Formatted: Font: Bold

Formatted: Indent: Left: 1.27 cm, Hanging: 1.27 cm, No bullets or numbering

Formatted: Font: Bold

Formatted: Font: 11 pt

Formatted: Font: 11 pt

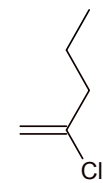
Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt, Bold

30

C

The chloro group is directly bonded to the double bond.

- The p orbital of Cl overlaps with the π orbital of the C=C bond.
- hence lone pair on Cl delocalises into the C=C bond.
- strengthening the C – Cl bond.

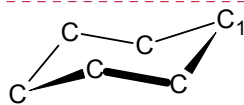
Hence, the Cl atom is resistant to nucleophilic substitution.

(v) Using the above information.

suggest a suitable reagent for step II2.

CH(CH₃)N₂

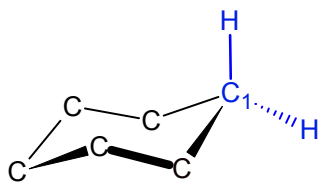
- (h) Unlike the cyclic alkanes smaller than cyclohexane, cyclohexane does not experience ring strain due to the “chair shape” arrangement adopted by the six carbons as seen below:



chair shape arrangement of carbon atoms

By copying out the chair shape arrangement above and drawing in the 3-D arrangement of hydrogen atoms bonded to C₁, explain why cyclohexane does not experience ring strain.

[22]



- Cyclohexane does not experience ring strain as cyclohexane bond angles are close to 109.5° due to the chair shape formation.

- (i) Explain the reaction with NaNH₂. [1]
- (ii) Hence, explain the formation of compound C. [1]
- (iii) Suggest skeletal structures for compounds B and C. [2]

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Space After: 0 pt, Line spacing: single

Formatted: Font: 10.5 pt

Formatted: Not Highlight

Formatted: Indent: Left: 1.27 cm, First line: 0 cm

Formatted: Font: Not Bold

Formatted: Not Highlight

Formatted: Indent: Left: 1.27 cm, No bullets or numbering, Tab stops: Not at 16.75 cm

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Font: 11 pt

Formatted: Indent: First line: 1.27 cm

Formatted: Indent: Hanging: 1.27 cm, Numbered + Level: 1 + Numbering Style: a, b, c, ... + Start at: 6 + Alignment: Left + Aligned at: 0.63 cm + Indent at: 1.27 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 1.25 cm, No bullets or numbering

Field Code Changed

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Indent: Left: 1.26 cm

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: English (Singapore)

When a current of 1.0 A was passed through aqueous potassium maleate ($\text{KO}_2\text{CCH}=\text{CHCO}_2\text{K}$) for 15 minutes, it was found that $110\text{ cm}^3\text{ H}_2$, measured at r.t.p., was collected at the cathode. The following reaction took place:



(d) State the relationship between the Faraday constant, F and the Avogadro's constant, L . [1]

(e) Using the data above and the *Data Booklet*, calculate a value for Avogadro's constant. [3]

(f) Ethyne and CO_2 gas were produced at the anode. In order to determine the stoichiometry of the anode reaction, the volume of the gases collected at the anode was measured. The anode gas was first passed through aqueous NaOH before being collected in a gas syringe. The following data was collected:

- mass of NaOH before experiment = 10.501 g
- mass of NaOH after experiment = 10.904 g
- initial reading on syringe = 10.0 cm^3
- final reading on syringe = 120.0 cm^3

(i) State the oxidation state of carbon in ethyne. [1]

(ii) With the help of an equation, explain the purpose of passing the anode gas through NaOH . [1]

(iii) Calculate the volume of CO_2 produced, assuming r.t.p. conditions. [1]

(iv) Hence, suggest an ionic equation for the reaction that occurred at the anode. [1]

(g) When aqueous potassium maleate was acidified, maleic acid, $\text{HO}_2\text{CCH}=\text{CHCO}_2\text{H}$ ($\text{p}K_{\text{a}1} = 1.90$ and $\text{p}K_{\text{a}2} = 6.07$) was liberated. Fumaric acid ($\text{p}K_{\text{a}1} = 3.03$ and $\text{p}K_{\text{a}2} = 4.44$) is a stereoisomer of maleic acid.

With a suitable illustration, suggest a reason why maleic acid has a lower $\text{p}K_{\text{a}1}$ but higher $\text{p}K_{\text{a}2}$ than fumaric acid.

[2]

[Total: 20]

Formatted: Font: Not Italic, Complex Script Font: Not Italic, Subscript

Formatted: Highlight

- 5(a) Dopamine is an organic compound of the catecholamine and phenethylamine families that plays several important roles in the brain and body. Its name is derived from its chemical structure: it is an amine synthesised by removing a carboxyl group from a molecule of its precursor compound, L-DOPA. The halogens and their compounds, show many similarities and trends in their properties. Some data are given for the elements fluorine, chlorine and iodine.

Element	Bond Energy / kJ mol ⁻¹	Standard enthalpy change of atomisation / kJ mol ⁻¹
Fluorine	158	79
Chlorine	242	121
Bromine	193	112
Iodine	151	107

(i) For fluorine and chlorine, their enthalpy changes of atomisation are half the value of their respective bond energies. For bromine and iodine, their enthalpy changes of atomisation are much more than half the value of their respective bond energies.

Explain in detail for this difference. [1]

Fluorine and chlorine are gases but bromine is a liquid and iodine is a solid at room temperature. The enthalpy change of atomisation includes the energy required to change Br₂(l) → Br₂(g) and I₂(s) → I₂(g) [to vapourise bromine and iodine to the gaseous state].

(ii) The standard enthalpy change of formation of iodine monochloride, I-Cl, is -24.0 kJ mol⁻¹.

Use this information and the data from the table above to calculate the I-Cl bond energy. [1]

$$\Delta H_f = \sum \text{Bonds broken} - \sum \text{Bonds formed}$$

$$-24.0 = \frac{1}{2} \text{BE}(\text{Cl}-\text{Cl}) + \frac{1}{2} \text{BE}(\text{I}-\text{I}) - \text{BE}(\text{I}-\text{Cl})$$

$$\text{BE}(\text{I}-\text{Cl}) = +220.5$$

$$= +221 \text{ kJ mol}^{-1}$$

Below is a synthetic route involving L-DOPA and dopamine:

Formatted: English (Singapore)

Formatted: Indent: Left: 0 cm, Hanging: 1.25 cm, No bullets or numbering

Formatted: Space Before: 3 pt, After: 3 pt

Formatted Table

Formatted: Space Before: 3 pt, After: 3 pt

Formatted: Space Before: 3 pt, After: 3 pt

Formatted: Space Before: 3 pt, After: 3 pt

Formatted: Space Before: 3 pt, After: 3 pt

Formatted: Font: Bold, Not Highlight

Formatted: Indent: Left: 1.27 cm, Hanging: 1.27 cm, Space After: 10 pt, Line spacing: Multiple 1.15 li, No bullets or numbering

Formatted: Not Highlight

Formatted: Highlight

Formatted: Indent: Left: 2.54 cm

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm, Space After: 0 pt, Line spacing: single

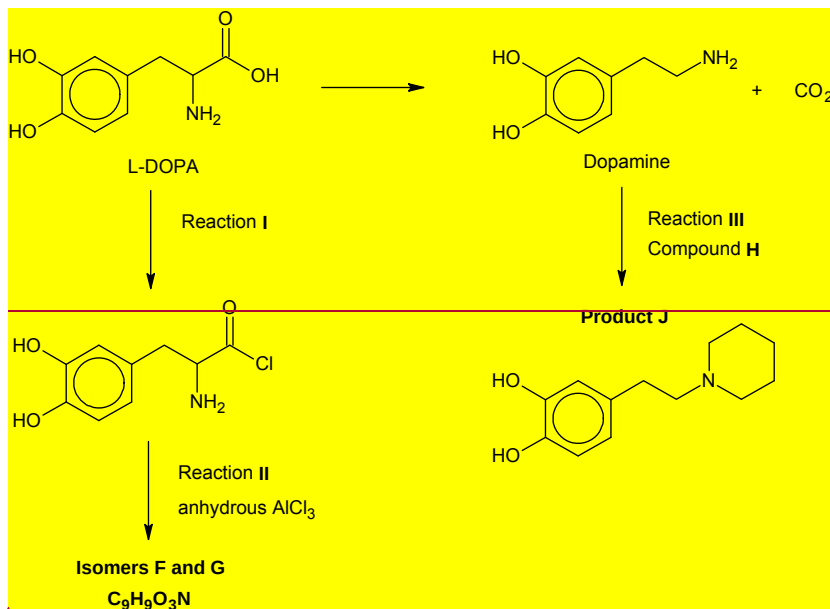
Formatted: Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Indent: Left: 2.54 cm, Space After: 0 pt, Line spacing: single

Formatted: Indent: First line: 0.63 cm

Formatted: Indent: Left: 3.17 cm

Formatted: Highlight



(i) State the reagents and conditions and any observations in Reaction I. [1]

(ii) Aluminium chloride is used as a catalyst in electrophilic substitution reactions. The chlorination of benzene is represented by the following overall equation.



The reaction occurs in several steps.

The first step is the reaction between Cl_2 and AlCl_3 .



The benzene ring is then attacked by the Cl^+ cation in the second step.

AlCl_3 reacts in a similar way with acyl chlorides, producing a carbocation that can then attack a benzene ring.

Predict the structures of isomers F and G in Reaction II. [2]

(iii) In Reaction III, dopamine was reacted with alkyl halide H to give the final product J. Draw the displayed formula of H. [2]

(iii) Explain why your answer in (ii) does not correspond to the average of the bond energies of I-I and Cl-Cl. [1]

I-Cl is polar while I-I and Cl-Cl are non-polar molecules. There is additional electrostatic attraction between $\text{I}^{\delta+}$ and $\text{Cl}^{\delta-}$. Hence, the actual bond energy of I-Cl is greater than the average bond energies of I-I and Cl-Cl.

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Indent: Left: 0 cm, First line: 0 cm

Formatted: Highlight

Formatted: Highlight

Formatted: Indent: Left: 0 cm, First line: 0 cm

Formatted: Indent: Left: 3.81 cm

Formatted: Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm

Formatted: Right: -0.01 cm, Don't adjust right indent when grid is defined, Space After: 0 pt, Line spacing: single, Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 2 + Alignment: Left + Aligned at: 1.27 cm + Indent at: 2.54 cm, Don't use Asian rules to control first and last character, Don't allow hanging punctuation, Don't adjust space between Latin and Asian text, Don't adjust space between Asian text and numbers

Formatted: Indent: Left: 2.54 cm, No bullets or numbering

(b) ICl reacts with pure water to form HCl and HI:

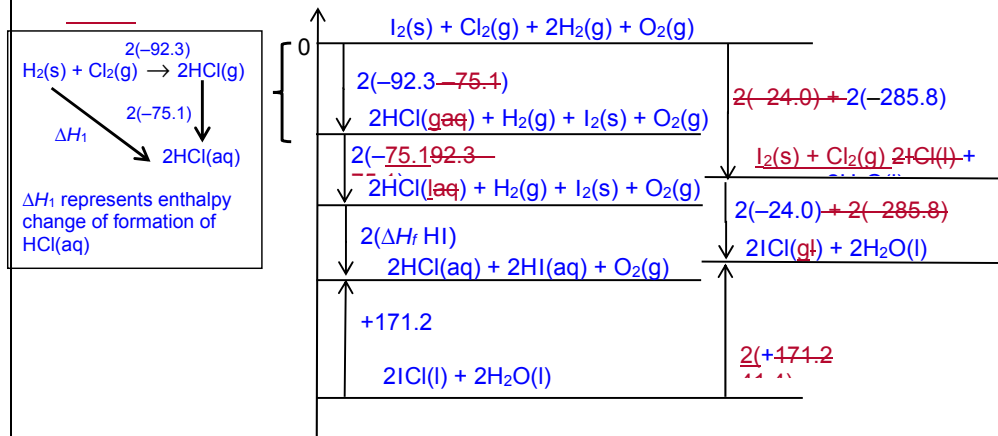


Using ΔH_r , the following data, as well as relevant data from a(ii), draw an energy level diagram to calculate the enthalpy change of formation of aqueous HI.

Label your diagram, and draw arrows representing the energy terms involved. Use words or symbol to represent these energy terms.

	$\Delta H / \text{kJ mol}^{-1}$
Standard enthalpy change of formation of H_2O	-285.8
Standard enthalpy change of formation of gaseous HCl	-92.3
Standard enthalpy change of reaction: $\text{HCl(g)} \rightarrow \text{HCl(aq)}$	-75.1
Standard enthalpy change of vaporisation of liquid ICl	+41.4

Energy kJ mol^{-1}



[1]x2 Balanced equations. State symbols and enthalpy for 3 reactions x 2

[1] Label energy terms on diagram, arrows correctly

[1] Recognise the enthalpy change of formation-Energy level diagram (energy axis, '0' at elements level, correct direction of arrows)

By Hess' Law:

$$2(-24.0) + 2(-285.8) + (+171.2) = 2(41.4) + 2(-92.3) + 2(-75.1) + 2\Delta H_f(\text{HI(aq)})$$

$$2\Delta H_f(\text{HI(aq)}) = -2(-75.1) - 2(-92.3) + 2(-285.8) + 2(-24.0) - 2(41.4) + (+171.2)$$

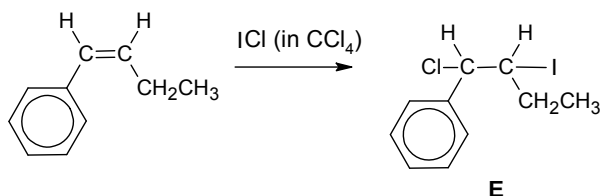
$$-48.0 - 571.6 + 171.2 + 334.8$$

$$\Delta H_f(\text{HI(aq)}) = -985.28 \text{ kJ mol}^{-1} \quad [1]$$

Formatted: Font: 11 pt, Complex Script Font: 11 pt, Highlight

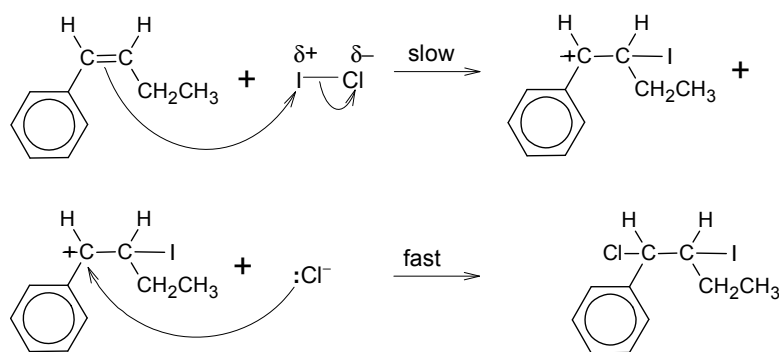
Formatted: Normal, Indent: Left: 0 cm, Hanging: 1.25 cm,
Space After: 0 pt, Line spacing: single, No bullets or
numbering, Tab stops: 1.25 cm, Left

(bc) ICl is a useful reagent in organic synthesis. It is used in the following reaction to form compound E.



(i) Describe the mechanism for the formation of E.

Electrophilic Addition



- Name of mechanism
- Correct arrows indicated
- Correct carbocation drawn
- Balanced equations
- Slow/ fast steps
- Charges on the atoms, lone pairs of electrons on Cl⁻

Every 2 points – 1 mark

(ii) Dopamine is a bidentate ligand. When different volumes of 0.0030 mol dm⁻³ of aqueous Cr(III) and 0.0020 mol dm⁻³ of alkaline dopamine solution were mixed, a complex R is formed. Analysis of R shows that its formula is [Cr(C₈H₉NO₂)_x(H₂O)_y]^{z+}, where x, y and z are integers.

To determine the stoichiometry of the complex ion formed, the colour intensities of these different mixtures were measured using a colorimeter. The following absorption spectrum was obtained.

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt, Not Highlight

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Centered

Formatted: List Paragraph, Indent: Left: 1.27 cm, Hanging: 1.27 cm, Tab stops: Not at 1.28 cm

Formatted: Highlight

Formatted: Indent: First line: 1.27 cm

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

Formatted: Indent: Left: 1.5 cm, First line: 1.23 cm

Formatted: Indent: Left: 1.27 cm, First line: 1.27 cm

Formatted: List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: 1 cm, Left + Not at 15.85 cm

Formatted: Not Highlight

Formatted: Indent: Left: 0 cm, First line: 0 cm

Formatted: Font: Bold, Not Highlight

Formatted: Font: Arial

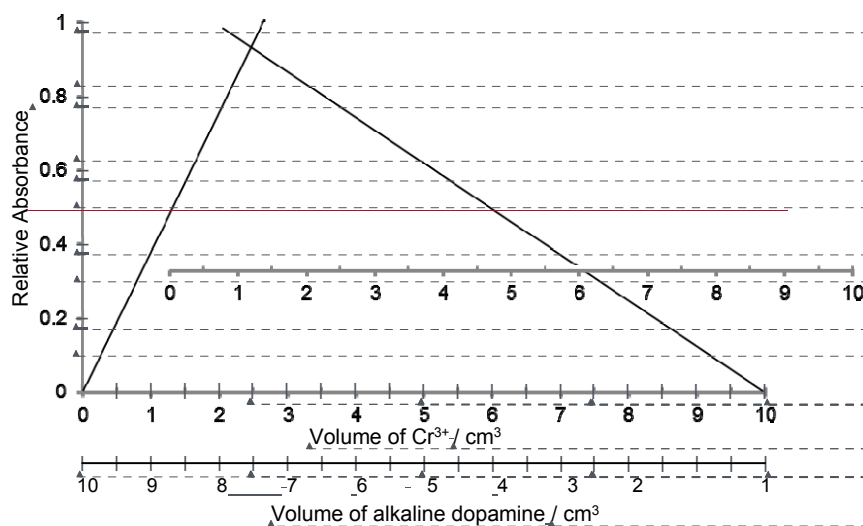
Formatted: Font: Arial

Formatted: Font: Not Bold

Formatted: Font: Not Bold

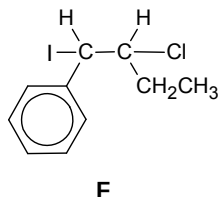
Formatted: Font: Arial

Formatted: Indent: Left: 0 cm

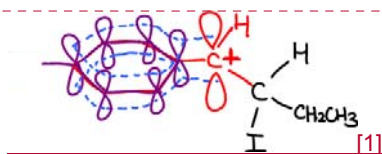


(i) Use the graph and the information given to determine the formula of complex **R**. Show your workings clearly. [3]

With the aid of a diagram, explain why **E** is formed and not **F**. [2]



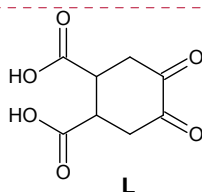
The carbocation and the C atoms in benzene ring are sp^2 hybridised. The (+) charge on carbocation is dispersed over the neighbouring benzene ring. Due to the effective overlap between the unhybridised p-orbitals of benzene and the empty p-orbital on the carbocation, the delocalised electrons makes the (+) charge on the carbocation less (+). [1]



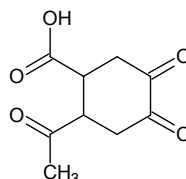
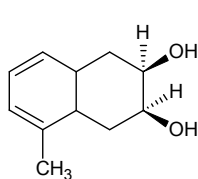
(d) Compound **J**, $C_{11}H_{16}O_2$, decolourises bromine water. 1 mole of **J** reacts with sodium metal to produce 22.7 dm^3 of hydrogen gas at s.t.p. On heating with acidified KMnO_4 , **K**, $C_9H_{10}O_5$, is the only organic product formed.

38

K reacts with sodium carbonate and 2,4-DNPH. When K reacts with alkaline aqueous iodine, L is formed upon acidification.



Suggest structures for J and K and explain the reactions described. [5]



Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt, Not Highlight

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: Font: 11 pt, Complex Script Font: 11 pt

Formatted: List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: 1 cm, Left + Not at 15.85 cm

Observations	Deductions
<u>J, C₁₁H₁₆O₂ decolourises bromine water</u>	<u>J could be alkene or phenol.</u> <u>An alkene undergoes electrophilic addition reaction with Br₂(aq) to form halogenoalkane.</u>
<u>22.7 dm³ of hydrogen is formed at s.t.p when J is reacted with sodium metal</u>	$n(\text{H}_2) = \frac{22.7}{22.7} = 1 \text{ mol of H}_2 \text{ is formed.}$ $\text{ROH} + \text{Na} \rightarrow \text{RO-Na}^+ + \frac{1}{2}\text{H}_2$ <u>Since 1 FG produces ½ mole of H₂, there must be 2 –OH groups present. –COOH group is absent as it will only produce only ½ mole of H₂ and 2 –COOH groups or 1 –OH & 1 –COOH groups cannot be present as it will not correspond to the molecular formula of J.</u>
<u>On heating with acidified KMnO₄, K, C₉H₁₀O₅, is the only organic product formed</u>	<u>Secondary alcohol and the alkene in J undergoes oxidation with KMnO₄ to form ketones and carboxylic acid.</u> <u>There is a decrease in 2C atoms – which suggest that ethane-1,2-dioic acid was oxidised to form CO₂.</u> <u>From the given structure of L, it can be seen that K is unlikely a phenol and should not contain benzene as there will not be oxidation that leads to a reduction of 2 C atoms.</u>
<u>K reacts with sodium carbonate and 2,4 DNPH</u>	<u>K undergoes acid-base with Na₂CO₃ → –COCH₃ present.</u> <u>K undergoes condensation with 2,4 DNPH → –confirms presence of ketones.</u>
<u>K reacts with alkaline aqueous iodine to form L</u>	<u>K undergoes mild oxidation with alkaline I₂(aq) → –COCH₃ present.</u> <u>K undergoes acid-base reaction with alkali to form salt → –COOH present.</u>

1 mark for each correct structure

3 marks for explanation

5 points – 3 marks

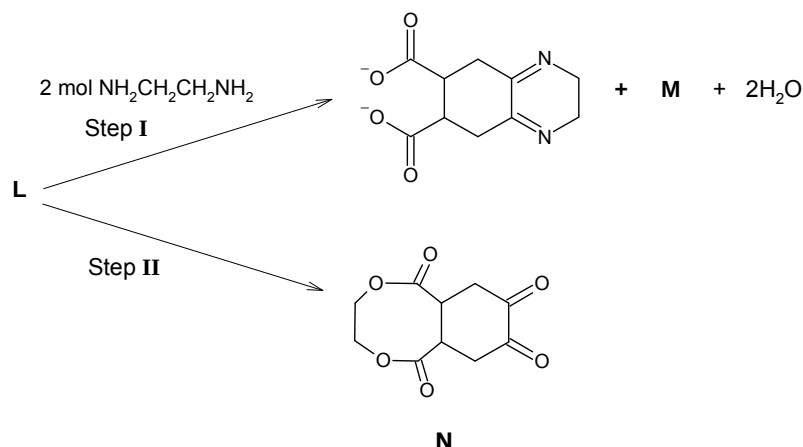
3 to 4 points – 2 marks

2 points – 1 mark

Formatted: Space After: 0 pt

Formatted: List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: 1 cm, Left + Not at 15.85 cm

(e) In the following reaction scheme, compounds M and N can be obtained from L.



Formatted: Indent: Hanging: 1.63 cm, Numbered + Level: 1 + Numbering Style: a, b, c, ... + Start at: 4 + Alignment: Left + Aligned at: 1 cm + Indent at: 1.63 cm

Formatted: Indent: Left: 1.63 cm, No bullets or numbering

(i) Draw the structure of M. State the type(s) of reaction in Step I. [2]

Formatted: Font: (Default) Times New Roman, Bold, Complex Script Font: Times New Roman

* $\text{NH}_3\text{CH}_2\text{CH}_2\text{NH}_3^+$ [1] Types of reaction: Condensation, Acid – Base [1]

(ii) Suggest reagents and conditions to synthesise product N from L. [1]

Formatted: Font: (Default) Arial, Complex Script Font: Arial

(iii) The crystal field describes the breaking of orbital degeneracy in transition metal complexes due to the presence of ligands. When the d-orbitals split into high energy and low energy orbitals, the difference in energy of the two levels is denoted as Δ_o . The relationship between Δ_o and colours of complexes can be described in the equation below:

$$\Delta_o = \frac{hc}{\lambda}$$

Formatted: List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: 1 cm, Left + Not at 15.85 cm

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

where h is Planck's constant, c is the speed of light and λ is the wavelength of light absorbed

colour	absorbed λ / nm
violet	410
indigo	430
blue	480
blue-green	500
green	530
yellow	580
orange	610
red	680

$\text{CH}_2(\text{OH})\text{CH}_2(\text{OH})$, concentrated H_2SO_4 heat under reflux.

OR

1) PCl_5 at rtp 2) $\text{CH}_2(\text{OH})\text{CH}_2(\text{OH})$, rtp

Given that Δ_e for complex **R** is 4.125×10^{-22} kJ and using relevant data from the *Data Booklet*, calculate the wavelength of light. Deduce the colour of complex **R**. [2]

Formatted: Highlight

(c) Iodine is not very soluble in water, it is freely soluble in KI(aq), according to the following equilibrium:



(i) Draw a fully labelled experimental set up for a voltaic cell made up of a $Cr_2O_7^{2-}/Cr^{3+}$ half-cell and a I_2/I^- half-cell under standard conditions. Indicate clearly the anode and cathode and show the flow of electrons. [3]

(ii) By using appropriate values from the *Data Booklet*, predict what, if anything, will happen when a small amount of acidified vanadium(II) chloride is added to the I_2/I^- half-cell. [3]

(d) Explain the following statements.

(i) BrF_3 is a covalent compound which exhibits electrical conductivity in liquid state at room temperature. With the aid of an equation, suggest an explanation for its electrical conductivity. [2]

(ii) $SiCl_4$ reacts violently in water but CCl_4 has no reaction with water. [1]

(iii) Compounds NeF_2 and NeF_4 do not exist but XeF_2 and XeF_4 exist. [1]

- [Total: 20]

Formatted: Right: -0.02 cm

Formatted: Right, Right: -0.02 cm

Formatted: Right: -0.02 cm

Formatted: Indent: Left: 0 cm, First line: 0 cm, Right: -0.02 cm

Formatted: Right: -0.02 cm

[End of Paper]

Formatted: Font: (Default) Arial, Complex Script Font: Arial

Page 7: [1] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [2] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:23:00 PM
Indent: Left: 1.27 cm, First line: 1.27 cm		
Page 7: [3] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt		
Page 7: [4] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [5] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [6] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [7] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [8] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 7: [9] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Complex Script Font: 11 pt		
Page 7: [10] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 12 pt		
Page 7: [11] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [12] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Indent: Left: 1.27 cm, First line: 1.27 cm		
Page 7: [13] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [14] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [15] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [16] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [17] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [18] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [19] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [20] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
Font: 11 pt		
Page 7: [21] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM

Indent: Left: 2.54 cm

Page 7: [22] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [23] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [24] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Indent: Left: 1.27 cm, First line: 1.27 cm

Page 7: [25] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Indent: Left: 2.54 cm

Page 7: [26] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [27] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [28] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [29] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 7: [30] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:24:00 PM
------------------------	---------------------	------------------------

English (United States)

Page 7: [31] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:45:00 PM
------------------------	---------------------	------------------------

Right: -0.07 cm

Page 7: [32] Formatted	Eunice Sim Kwei Yen	05/08/2018 04:15:00 PM
------------------------	---------------------	------------------------

Font: Bold

Page 7: [33] Formatted	Eunice Sim Kwei Yen	05/08/2018 04:15:00 PM
------------------------	---------------------	------------------------

Font: Bold

Page 7: [34] Formatted	Eunice Sim Kwei Yen	05/08/2018 04:15:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Bold, Complex Script Font: 11 pt

Page 7: [35] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:45:00 PM
------------------------	---------------------	------------------------

Right: -0.07 cm

Page 7: [36] Formatted	Eunice Sim Kwei Yen	05/08/2018 04:15:00 PM
------------------------	---------------------	------------------------

Font: Bold

Page 7: [37] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [38] Formatted	Eunice Sim Kwei Yen	05/08/2018 04:03:00 PM
------------------------	---------------------	------------------------

Indent: First line: 0 cm, Right: -0.07 cm

Page 7: [39] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [40] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Bold, Complex Script Font: 11 pt, Bold, (Complex) Arabic (Saudi Arabia), English (United States), Do not check spelling or grammar

Page 7: [41] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [42] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [43] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [44] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [45] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [46] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [47] Formatted	Eunice Sim Kwei Yen	15/08/2018 03:28:00 PM
------------------------	---------------------	------------------------

Centered

Page 7: [48] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 7: [49] Formatted	Eunice Sim Kwei Yen	15/08/2018 03:37:00 PM
------------------------	---------------------	------------------------

Numbered + Level: 1 + Numbering Style: i, ii, iii, ... + Start at: 1 + Alignment: Left +
Aligned at: 1.27 cm + Indent at: 2.54 cm

Page 7: [50] Formatted	Eunice Sim Kwei Yen	05/08/2018 03:51:00 PM
------------------------	---------------------	------------------------

Font: 11 pt

Page 37: [51] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

No bullets or numbering, Tab stops: Not at 1 cm

Page 37: [52] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

Font: Not Bold, Complex Script Font: Arial

Page 37: [53] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

Complex Script Font: Arial

Page 37: [54] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

Font: Arial

Page 37: [55] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control,
Keep lines together, Tab stops: Not at 15.85 cm

Page 37: [56] Formatted	Theresa Line Ishak	13/08/2018 08:08:00 AM
-------------------------	--------------------	------------------------

Font: Arial

Page 37: [57] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

Font: Arial

Page 37: [58] Formatted	Theresa Line Ishak	13/08/2018 08:08:00 AM
-------------------------	--------------------	------------------------

Font: Arial

Page 37: [59] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
-------------------------	---------------------	------------------------

Font: Arial

Page 37: [60] Formatted	Theresa Line Ishak	13/08/2018 07:51:00 AM
-------------------------	--------------------	------------------------

List Paragraph, Centered, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control,
Keep lines together, Tab stops: Not at 15.85 cm

Page 37: [61] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:59:00 PM
Font: 11 pt		
Page 37: [62] Formatted	Eunice Sim Kwei Yen	14/09/2018 01:59:00 PM
Font: 11 pt		
Page 37: [63] Formatted	Theresia Line Ishak	13/08/2018 07:51:00 AM
List Paragraph, Centered, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: Not at 15.85 cm		
Page 37: [64] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
Font: Arial		
Page 37: [65] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:00:00 AM
List Paragraph, Justified, Indent: Left: 0 cm, Right: 0.17 cm, No widow/orphan control, Keep lines together, Tab stops: 1 cm, Left + Not at 15.85 cm		
Page 37: [66] Formatted	Eunice Sim Kwei Yen	06/08/2018 12:02:00 AM
Font: Arial		
Page 37: [67] Formatted	Theresia Line Ishak	13/08/2018 08:22:00 AM
Font: (Default) Arial, Complex Script Font: Arial		
Page 37: [68] Formatted	Theresia Line Ishak	24/08/2016 07:20:00 PM
Font: 11 pt, Complex Script Font: 11 pt		
Page 37: [69] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold		
Page 37: [70] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [71] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold		
Page 37: [72] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [73] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold		
Page 37: [74] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [75] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold		
Page 37: [76] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [77] Formatted	Theresia Line Ishak	24/08/2016 07:04:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [78] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
Font: Bold, Complex Script Font: Bold, English (Singapore)		
Page 37: [79] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
Font: Bold, Complex Script Font: Bold		
Page 37: [80] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [81] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [82] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [83] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [84] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [85] Formatted	Theresia Line Ishak	24/08/2016 07:20:00 PM
-------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 37: [86] Formatted	Theresia Line Ishak	24/08/2016 07:20:00 PM
-------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 37: [87] Formatted	Theresia Line Ishak	24/08/2016 07:20:00 PM
-------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 37: [88] Formatted	Theresia Line Ishak	24/08/2016 07:20:00 PM
-------------------------	---------------------	------------------------

Font: 11 pt, Complex Script Font: 11 pt

Page 37: [89] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [90] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [91] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [92] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [93] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [94] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)

Page 37: [95] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold

Page 37: [96] Formatted	Theresia Line Ishak	24/08/2016 06:52:00 PM
-------------------------	---------------------	------------------------

Font: Bold, Complex Script Font: Bold, English (Singapore)